

Bachelor's Thesis Assignment



162826

Institut: Department of Intelligent Systems (DITS)
Student: **Baraniuk Nestor**
Programme: Information Technology
Title: **Application of the ExpectiMiniMax procedure to the Backgammon.**
Category: Artificial Intelligence
Academic year: 2024/25

Assignment:

1. Study the rules of the Backgammon game and the ExpectiMiniMax procedure.
2. Design a program to play this game with the given procedure.
3. Design functions for evaluating game states.
4. Experiment with the designed evaluation functions.
5. Evaluate the results achieved.

Literature:

- Russel, S., Norvig, P.: Artificial Intelligence, A modern Approach, third edition, Prentice Hall, 2010
- Simborg, P.: The Rules of Backgammon, www.WycliffeBrothers.com, 2011
- Melko E., Nagy B.: Optimal strategy in games with chance nodes, Acta Cybernetica 18, 2007

Requirements for the semestral defence:

The first two points of assignment.

Detailed formal requirements can be found at <https://www.fit.vut.cz/study/theses/>

Supervisor: **Zbořil František, doc. Ing., CSc.**
Head of Department: Kočí Radek, Ing., Ph.D.
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